DOCUMENT REGISTRY ID: GEO-BK-142

Borehole Exploration Report â€” BK-142

Core Drilling Log, Lithological Interpretation & Reserve Estimation

CATEGORY	MINE / BLOCK	YEAR	CLASSIFICATION
Geological	Barkakana Block	2025	RESTRICTED / VERIFIED

DOCUMENT VERIFICATION & ACCESS REGISTER

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SECTION 1.0 â€” DOCUMENT INFORMATION & CONTROL

Official Technical Record & Geological Data Logging

This document represents the certified technical record for Borehole BK-142, executed in the Barkakana Block under the Damodar Valley Geological Exploration Program.

All drill core samples and subsequent geophysical logs have been processed in accordance with Indian Standard Specifications IS:4410 and Directorate General of Mines Safety (DGMS) norms.

Record Control Registry

Parameter	Specification / Value	Verification Agency	Status
Borehole ID	BK-142 (Core Diameter: NQ/HQ)	Geological Services Cell	Certified
Block Name	Barkakana Coal Block, Sector IV	CMPDI Regional Inst.	Approved
Total Drilled Depth	342.50 Meters (True Vertical)	Core Logging Division	Verified
Rig Type	Longyear 44 Hydraulic Wireline	Exploration Drilling Ltd	Operational
Spud Date	12 November 2024 (08:00 IST)	Field Shift Log A	Logged
Completion Date	18 December 2024 (17:30 IST)	Final Borehole Survey	Completed

VERIFIED PARAMETER REGISTRATION:

â€¢ Cumulative Core Recovery:	94.6%	Exceeds 90% threshold
â€¢ Geophysical Wireline Log:	Complete (340m)	Gamma, Density, Caliper

SECTION 2.0 â€” EXECUTIVE SUMMARY & KEY FINDINGS

Official Technical Record & Geological Data Logging

Borehole BK-142 intersected four distinct coal-bearing seams belonging to the Barakar Formation of Lower Gondwana age.

The cumulative net coal thickness encountered is 14.85 meters across the stratigraphic column from depth 64.20m to 318.40m.

Proximate analysis confirms premium semi-coking to non-coking industrial grades ranging from Grade G-3 to G-5 with mean Gross Calorific Value (GCV) of 5,420 kcal/kg.

CRITICAL TECHNICAL OBSERVATION & DIRECTIVE:

Coal Seam IX Top exhibited zero structural shearing with excellent floor sandstone competency, rendering it prime for highwall mining.

Executive Coal Seam Intersection Summary

Seam Name	Floor Depth (m)	Thickness (m)	GCV (kcal/kg)	Ash %	Grade
Seam IX Top	68.40	3.20	5,680	19.4%	G-3
Seam VIII	142.10	4.15	5,410	22.8%	G-4
Seam VII Lower	228.60	2.80	5,290	24.1%	G-5
Seam V Composite	314.20	4.70	5,300	23.5%	G-5

SECTION 3.0 â€” PROJECT LOCATION & TOPOGRAPHICAL SETTING

Official Technical Record & Geological Data Logging

The Barkakana Block is located in the southeastern quadrant of the Ramgarh Coalfield. Topography is characterized by gently undulating terrain with an average elevation of 312m above Mean Sea Level (MSL).

The exploration area is bounded by latitudes 23 36'40"N to 23 38'15"N and longitudes 85 28'30"E to 85 31'10"E, covering an aggregate concession area of 14.6 square kilometers.

Boundary Coordinates and Lease Limits

Pillar ID	Easting (UTM 45N)	Northing (UTM 45N)	RL (m MSL)	Remarks
P-01	344512.45 m	2614890.12 m	314.50 m	Northern Corner
P-02	346890.30 m	2614820.50 m	312.80 m	Eastern Corner
P-03	346740.15 m	2611950.80 m	308.20 m	Southern Corner
P-04	344380.90 m	2612010.40 m	310.40 m	Western Corner

SECTION 4.0 â€” REGIONAL GEOLOGICAL SETTING & BASIN STRUCTURE

Official Technical Record & Geological Data Logging

The Damodar Valley Basin system forms an east-west trending linear intracratonic rift basin within the Chhotanagpur Granite Gneiss Complex (CGGC).

Sedimentation initiated with glacial Talchir tillites followed by extensive fluvial and deltaic deposits of the coal-rich Barakar Formation.

VERIFIED PARAMETER REGISTRATION:

â€¢ **Basin Tectonic Style:**	`Half-Graben with Northern Fault`	Strike: N75W
â€¢ **Dip of Strata:**	`6 to 8 deg towards S20W`	Uniform structural dip
â€¢ **Basement Contact:**	`Precambrian Metamorphics at 880m`	Unconformable

SECTION 5.0 â€” EXPLORATION OBJECTIVES & DRILLING METHODOLOGY

Official Technical Record & Geological Data Logging

The objective of BK-142 was to delineate deep seam continuity beneath the alluvial cover and assess the coking characteristics of Seam VIII and IX.

Drilling was conducted using continuous wireline core retrieval with synthetic polymer mud flush to maintain borehole integrity and prevent seam washouts.

Drilling Tooling and Fluid Specifications

Depth Interval (m)	Casing / Bit Size	Mud Weight (SG)	Viscosity (s)	Core Barrel
0.00 - 18.00	PW Surface Casing / 146mm	1.05	35	Single Tube
18.00 - 140.00	HQ Wireline / 96mm Bit	1.08	42	HQ-3 Triple Tube
140.00 - 342.50	NQ Wireline / 75.7mm Bit	1.10	45	NQ-3 Split Tube

SECTION 6.0 â€” BOREHOLE COORDINATES & TECHNICAL SURVEY DATA

Official Technical Record & Geological Data Logging

Differential Global Positioning System (DGPS) survey was performed using Leica Viva GS15 receivers tied to the Survey of India Great Trigonometrical Survey (GTS) bench mark.

Certified Spatial & Geodetic Data

Parameter	Instrument / Reference	Measured Value	Accuracy Tolerance
Latitude (WGS84)	DGPS Dual Frequency	23 37' 14.821" N	+/- 5 mm
Longitude (WGS84)	DGPS Dual Frequency	85 29' 48.330" E	+/- 5 mm
Collar Elevation (RL)	Auto Level Survey	312.450 m MSL	+/- 2 mm
Total Depth Drilled	Steel Tape Calibrated	342.500 m	+/- 0.05 m
Borehole Verticality	Multi-Shot Gyro Tool	0.42 deg max deviation	DGMS Approved

SECTION 7.0 â€” BOREHOLE DRILLING SUMMARY & OPERATIONAL CHRONOLOGY

Official Technical Record & Geological Data Logging

Drilling operations commenced on 12 November 2024 and reached the planned target depth of 342.50m without catastrophic loss of circulation or borehole wall collapse.

Progress Summary by Shift & Penetration Rate

Drilling Phase	Interval (m)	Drilling Hours	Rate of Pen. (m/hr)	Core Rec. %
Surface Alluvium	0.00 - 18.00	14.5 hrs	1.24 m/hr	78.2%
Upper Barakar	18.00 - 150.00	86.0 hrs	1.53 m/hr	96.4%
Middle Barakar	150.00 - 275.00	98.5 hrs	1.27 m/hr	95.8%
Lower Barakar	275.00 - 342.50	62.0 hrs	1.09 m/hr	97.1%

SECTION 8.0 â€” LITHOLOGICAL DESCRIPTION & DETAILED CORE LOGGING

Official Technical Record & Geological Data Logging

The borehole intersected alternating cycles of coarse to medium-grained arkosic sandstones, siltstones, carbonaceous shales, and high-rank bituminous coal seams.

Sideritic nodules and calcareous cementation were noted between 180m and 210m depth.

Macro-Lithological Unit Breakdown

Depth From (m)	Depth To (m)	Lithology Code	Description & Texture	RQD %
0.00	18.20	SOIL/ALV	Recent soil, clayey silt and gravel	0%
18.20	64.20	SST-CRS	Coarse feldspathic sandstone, cross-bedded	82%
64.20	68.40	COAL-IX	Banded bright coal, vitreous luster	94%
68.40	137.95	SST-MED	Medium grey sandstone with shale partings	88%
137.95	142.10	COAL-VIII	Semi-bright coal with clarain bands	96%
142.10	225.80	SH-CARB	Dark grey laminated carbonaceous shale	91%

SECTION 9.0 â€” STRATIGRAPHIC SEQUENCE & DEPOSITIONAL FACIES

Official Technical Record & Geological Data Logging

The encountered sequence represents a classic fining-upward fluvial cyclothem characteristic of channel-fill, point bar, and overbank swamp environments in the Barakar formation.

VERIFIED PARAMETER REGISTRATION:

â€¢ **Dominant Depositional Regime:**	**Braided to Meandering River Floodplain**	Fluvial System
â€¢ **Paleo-Current Direction:**	**NW to SE (135 deg Azimuth)**	Cross-bedding dip
â€¢ **Volumetric Sand:Shale Ratio:**	**3.4 : 1**	Sand-dominated Barakar unit

SECTION 10.0 â€” COAL SEAM INTERSECTIONS & BOREHOLE CORRELATION

Official Technical Record & Geological Data Logging

Four economic coal horizons were intersected. Correlative ties with adjacent boreholes BK-140 and BK-141 indicate high lateral continuity across a 2.4 km strike length.

Detailed Seam Correlation Register

Seam ID	Roof Depth (m)	Floor Depth (m)	Net Coal (m)	Dirt Bands (m)	Dip (deg)
Seam IX Top	64.20	68.40	3.20	0.00 (Clean)	7.2 deg
Seam VIII	137.95	142.10	4.15	0.20 (Carbonaceous)	6.8 deg
Seam VII Lower	225.80	228.60	2.80	0.15 (Shale)	7.0 deg
Seam V Composite	309.50	314.20	4.70	0.35 (Siltstone)	6.5 deg

SECTION 11.0 â€” DEPTH AND THICKNESS DETAILED TABULATION

Official Technical Record & Geological Data Logging

All seam intersections were verified using dual-detector density wireline logs (Short Space & Long Space Density) to establish millimeter-precise coal-rock contacts.

Core Thickness vs Geophysical Density Thickness

Seam ID	Drill Core Thk (m)	Geophys Thk (m)	True Vert Thk (m)	Variance (mm)
Seam IX Top	3.20 m	3.22 m	3.17 m	+20 mm
Seam VIII	4.15 m	4.14 m	4.11 m	-10 mm
Seam VII Lower	2.80 m	2.83 m	2.78 m	+30 mm
Seam V Composite	4.70 m	4.68 m	4.66 m	-20 mm

SECTION 12.0 â€” COAL QUALITY PARAMETERS & LABORATORY ASSAY

Official Technical Record & Geological Data Logging

Sample cores were wrapped in plastic seal immediately upon extraction and dispatched to the CMPDI Regional Laboratory for proximate and ultimate chemical analyses.

VERIFIED PARAMETER REGISTRATION:

â€¢ **Crucible Swelling Number (CSN):**	1.5 to 2.5	Semi-coking characteristics
â€¢ **Total Sulphur Content:**	0.42% to 0.58%	Low sulphur environment
â€¢ **Hardgrove Grindability Index (HGI):**	62 to 68	Easy pulverization

SECTION 13.0 â€” ASH, MOISTURE, VOLATILE MATTER & FIXED CARBON

Official Technical Record & Geological Data Logging

Proximate analysis results conducted at 60% Relative Humidity and 40 deg C temperature according to IS:1350 (Part-I) standards.

Seam-Wise Proximate Analysis Results

Seam ID	Moisture %	Ash %	Volatile Matter %	Fixed Carbon %	Gross CV (kcal/kg)
Seam IX Top	4.8%	19.4%	28.2%	47.6%	5,680
Seam VIII	4.2%	22.8%	26.5%	46.5%	5,410
Seam VII Lower	3.9%	24.1%	25.0%	47.0%	5,290
Seam V Composite	3.5%	23.5%	25.8%	47.2%	5,300

SECTION 14.0 â€” GEOLOGICAL INTERPRETATION & STRUCTURAL TECTONICS

Official Technical Record & Geological Data Logging

The block shows remarkable structural stability. No intra-block major normal faults with throw exceeding 5 meters were identified in the BK-142 borehole radius.

Joint sets observed in sandstone roof strata exhibit steep dips (75 to 85 deg) with average joint spacing of 1.2m.

CRITICAL TECHNICAL OBSERVATION & DIRECTIVE:

Absence of basic igneous intrusives confirms zero thermal alteration (Jhama) in the coal seams.

SECTION 15.0 â€” STRUCTURAL OBSERVATIONS & GEOMECHANICAL PROPERTIES

Official Technical Record & Geological Data Logging

Point load testing and Unconfined Compressive Strength (UCS) tests were executed on drill cores from roof and floor strata.

Geomechanical Rock Mass Parameters

Lithology Unit	Depth Interval (m)	UCS (MPa)	Tensile Str. (MPa)	Youngs Modulus (GPa)
Roof Sandstone	58.00 - 64.20	48.5 MPa	4.2 MPa	18.4 GPa
Seam IX Coal	64.20 - 68.40	18.2 MPa	1.8 MPa	3.1 GPa
Floor Shale	68.40 - 74.00	32.0 MPa	2.6 MPa	11.8 GPa
Deep Sandstone	300.00 - 309.50	62.4 MPa	5.1 MPa	24.2 GPa

SECTION 16.0 â€” RESERVE IMPLICATIONS & MINERAL INVENTORY

Official Technical Record & Geological Data Logging

Using an influence radius of 200m around BK-142, geological reserves have been estimated under the United Nations Framework Classification (UNFC) System Category 111 (Proved).

Proved Geological Reserve Estimates (BK-142 Influence Area)

Seam ID	Influence Area (sq.m)	Net Thickness (m)	Specific Gravity	Proved Reserve (MT)
Seam IX Top	125,600 sq.m	3.20 m	1.42	0.571 MT
Seam VIII	125,600 sq.m	4.15 m	1.44	0.751 MT
Seam VII Lower	125,600 sq.m	2.80 m	1.45	0.510 MT
Seam V Composite	125,600 sq.m	4.70 m	1.45	0.856 MT
TOTAL BLOCK	125,600 sq.m	14.85 m	1.44	2.688 MT

SECTION 17.0 â€” HYDROGEOLOGICAL REGIME & AQUIFER CHARACTERISTICS

Official Technical Record & Geological Data Logging

Piezometric monitoring conducted post-completion recorded the static water level at 8.40m below ground level.

Permeability testing (packer tests) in the sandstone horizon between 68m and 137m indicates low hydraulic conductivity (K = 0.042 m/day).

VERIFIED PARAMETER REGISTRATION:

â€¢ Static Water Level:	`8.40 m bgl`	Post-monsoon stabilized
â€¢ Aquifer Inflow Rate:	`1.8 liters/sec`	Manageable for opencast dewatering
â€¢ Water Quality (pH):	`7.2`	Non-acidic, potable baseline

SECTION 18.0 â€” DATA QUALITY NOTES & CORE RECOVERY VERIFICATION

Official Technical Record & Geological Data Logging

Strict quality assurance was maintained. The core recovery exceeded 95% in all coal seam horizons, meeting the mandatory requirement for coal resource categorization.

Core Recovery Quality Assurance Verification

Run Number	Drilled Run (m)	Recovered Core (m)	Recovery %	Quality Inspector
Run #22 (Seam IX)	3.50 m	3.42 m	97.7%	S. Mahato (Senior Geologist)
Run #48 (Seam VIII)	4.50 m	4.40 m	97.8%	S. Mahato (Senior Geologist)
Run #76 (Seam VII)	3.00 m	2.95 m	98.3%	A. Roy (Exploration Officer)
Run #99 (Seam V)	5.00 m	4.85 m	97.0%	A. Roy (Exploration Officer)

SECTION 19.0 â€” RECOMMENDATIONS FOR MINE PLANNING & FEASIBILITY

Official Technical Record & Geological Data Logging

1. Seam IX Top is recommended for initial box-cut excavation due to its shallow depth (64m) and favorable stripping ratio of 4.2 : 1.

2. Sandstone roof strata exhibit high compressive strength and require systematic pre-splitting or deep-hole blasting for optimal fragmentation.

3. Infill drilling at 200m grid spacing is advised to confirm pinch-out behavior of Seam VII towards the eastern boundary.

SECTION 20.0 â€” CONCLUSION & OFFICIAL CERTIFICATION

Official Technical Record & Geological Data Logging

Borehole BK-142 has successfully validated the multi-seam coal potential of Barkakana Block Sector IV, adding 2.688 MT of Proved Geological Reserves to the Central Repository.

The data presented herein has been verified against digital geophysical logs and laboratory assay certificates and is recommended for inclusion in the Master Mining Plan.

CRITICAL TECHNICAL OBSERVATION & DIRECTIVE:

Certified and digitally archived under Coal Intelligence Repository ID: GEO-BK-142.

VERIFIED PARAMETER REGISTRATION:

â€¢ Lead Geologist Sign-Off:	Sunita Mahato, M.Sc (App. Geol.)	Certified on 14-Feb-2025
â€¢ Chief Mining Officer:	Dr. Rajeshwar Sharma, F.I.E.	Approved on 15-Feb-2025

SECTION 21.0 â€” APPENDIX: GEOPHYSICAL LOG CURVES & LABORATORY CERTS

Official Technical Record & Geological Data Logging

CMPDI Regional Geochemical Laboratory Assay Certificate No. CMPDI/LAB/2025/GEO-8819.

Wireline Geophysical Log Multi-Channel Calibration Certificate No. LOG-WIR-2024-411.

Survey Bench Mark Reference: GTS Station Ramgarh No. 412 (RL: 312.45m).

Laboratory Assay Certification Numbers

Sample Core ID	CMPDI Assay Lab No.	Testing Date	Certifying Chemist
BK142-S9-C1	LAB-2025-0814	22-Jan-2025	Dr. P. N. Mukherjee
BK142-S8-C2	LAB-2025-0815	23-Jan-2025	Dr. P. N. Mukherjee
BK142-S7-C3	LAB-2025-0816	24-Jan-2025	M. Sengupta
BK142-S5-C4	LAB-2025-0817	25-Jan-2025	M. Sengupta